

Yoga for Combating Covid - 19 Pandemic - A Review

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ABSTRACT

Yoga has been accepted all over the world for its health-promoting and wellness-creating aspects. For fighting this Covid-19 Pandemic the immunity of the host is very important. There is various clinical evidence of the efficacy of yoga against Communicable disease settings and also for non-communicable Clinical Conditions. Thus, yoga has a holistic approach to strengthening our physical as well as mental health. But we must also understand that yoga is not an alternative to medicine but a prevention to stay away from disease as much as possible.

KEYWORDS: Covid-19, Yoga, Pandemic, Pranayama, Asana

INTRODUCTION

The spread of COVID-19 disease and the catastrophe we are all encountering today makes us search for answers and course improvements urgently craved for the very survival of humans. This current devastating pandemic had left many doubts on our knowledge base along with individual and social ways of life.

For the eradication of any infection, the immunity of the host is of utmost importance. Disturbed immune systems recognized as elevated C-reactive protein levels and lymphopenia are recognized to be the characteristic feature in severely affected cases of COVID-19 infection.

The primary organ affected by novel coronavirus is lungs, but impact on various body systems has been reported. Pneumonia, lung fibrosis with ground

glass opacities, thickening of Broncho vascular bundles, hydrothorax and chest distress are the most common changes in respiratory system (Cao et al., 2020). Among other systems most common changes are seen in cardiac system, kidneys, gastrointestinal tract. In addition, a wide range of neurological and musculoskeletal symptoms are also seen. These symptoms range from myocarditis, acute myocardial infarction, pyroptosis in kidneys, anorexia, vomiting, nausea and abdominal pain to gastrointestinal bleed (Balachandar et al, 2020). Apart from medical impairments psychological and cognitive impairments may also be seen in COVID19 recovered patients. An Italian study found that 80 percent of patients admitted to COVID-

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19 rehabilitation unit had abnormal scores at cognitive assessment (Iannaccone et al, 2020). Social strain or the eventual neurotropic effects of the virus may also trigger onset of psychiatric symptoms (Orsini et al, 2020)

If our immune system is strong, we gave a decent chance of winning over this novel corona virus attack. Thus, the outcome of this infection would be either a state of immune suppression that debilitated the system leading to progression to severe damage or a welcome result of a positive immune response that combats COVID-19 as observed in recovering patients with mild symptoms.

Exercise has been considered as a main component of post COVID-19 rehabilitation. The first component included in majority of the guidelines is pulmonary rehabilitation to improve respiratory function. It includes positioning for breathlessness, strategies to improve chest expansion including thorax mobilizing exercises, diaphragmatic breathing and inspiratory muscle training.

Yoga has been welcomed all over the world for its health-promoting and wellness creating aspects. Various researches over three decades at various places of the world has shown the efficacy of yogic practices including meditation in establishing homeostasis in no communicable diseases by promoting a healthy lifestyle and reducing stress

levels. Yoga for correcting the imbalances of the body and seeking the interconnections between the mind and the body had been intensively investigated to provide a holistic framework for the betterment of the health of individuals.

Yoga as an exercise method may be one of the best strategies to achieve all these benefits. The combination of asana and pranayama (breathing techniques) may be highly beneficial for the patients in first 6 weeks before they can return to recommended levels of regular aerobic and strength training.

The objective of this review work is to know the effectiveness of yogic practices in the prevalent pandemic situation with scientific evidence.

REVIEW OF LITERATURE

Integrated Yoga for the Management of Non-communicable Clinical Conditions: Yoga or meditation facilitates the coordination among the set of homeostatic responses implicating the interaction among the endocrine, immune and nervous systems which is inferred by some available evidence (Gard T et al., 2014). Recent explanations of yoga state it as a comprehensive skill set of synergistic process tools that aid in directional feedback and modulation of the autonomic nervous system which outputs through integration between the central nervous system (CNS) and afferent and re-afferent inputs from

interception processes such as the chemosensory, viscerosensory and somatosensory (Gard T et al., 2014). The integrative system of yoga techniques comprised of postures (Sanskrit: asana), breath regulation (Sanskrit: pranayama), and meditation along with the conceptual corrections which could promote physical as well as mental wellbeing. Different effects are purported by these postures or asana, some develop concentration and coordination, some are stimulatory to the circulatory and nervous systems while others have a calming effect on the body. Various poses like corpse poses are used for elongated periods of relaxation.

In chronically stressed people with anxiety and depression (Telles S et al., 2007, Cramer H et al., 2018, Telles S et al., 2009), for acute stress in post-traumatic stress disorder, and many non-communicable diseases such as hypertension (Nivethitha L et al., 2016, Hagins M et al., 2013), asthma (Nagarathna R et al., 1985, Cramer H et al., 2014), diabetes (Nagarathna R et al., 2019, Raveendran AV et al., 2018, Innes KE et al., 2016, McDermott KA et al., 2014, Singh AK et al., 2019) and heart disease these therapeutic techniques of yoga have been reported to be beneficial clinically. Yoga has been reported repeatedly to facilitate the attainment of glycemic control and mitigate the influence of other risk factors associated with the complications in patients with diabetes as compared to control

conditions in particular. The abdominal pressure created during exhalation in Kapalabhati improves the efficacy of beta cells of the pancreas (Raveendran AV et al., 2018) as proposed by various experts. It can be further perceived as modulated interception or sensory modulation stimulated by the vigorous practice of Kapalabhati aids in the increased interception of the unusual glycemic control that is signaled through the sensory inputs of the CNS that in turn modulates the autonomic outputs to the pancreas and other different organs associated to disease pathophysiology.

Understanding from Clinical evidence in Efficacy of Yoga against Communicable Disease Settings:

Various beneficiary effects of yoga as an add-on strategy for the management of communicable diseases including tuberculosis (TB) (Visweswarai NK et al., 2004), influenza (Obasi CN et al., 2013; Barrett B et al., 2012) and human immunodeficiency virus (HIV) infection (Naoroibam R et al., 2016; Dunne EM et al., 2019; Hari Chandra BP et al., 2019) had various evidence whereas the status of the immune system is a significant factor that determines the advancement of the disease. Results from meditation or exercise for preventing acute respiratory illness trial (MEPARI) trial implied that meditation training evoked a larger reduction in global acute respiratory infection (ARI) severity as compared to exercise or the wait-list control

participants (Obasi CN et al., 2013; Barrett B et al., 2012). The findings of the study were found to be in concordance with previous literature on the beneficial effects of moderate-intensity exercise against the immune system and lessening the incidence of ARI illness (Obasi CN et al., 2013).

Likewise, a prospective randomized trial compared the effectiveness of two programs (yoga and breath awareness) as an add-on to anti-TB treatment in sputum-positive cases in a sanatorium in Bangalore (Visweswaraiah NK et al., 2004). The yoga group reflected a significantly promising reduction in symptom score and an increase in lung capacity and weight with an improved level of infection control and radiographic image as compared to the non-yoga group, at the end of 2 months. Noteworthy improvement in the psychological states as well as in the viral loads in patients suffering from HIV-1 infection (Naoroibam R et al., 2016) has been reported as a consequence of one month of integrated yoga (IY) intervention. Yoga has also been reported to be a beneficial intervention for development in psychological health and stress management among AIDS/HIV patients (Naoroibam R et al., 2016; Dunne EM et al., 2019; Hari Chandra BP et al., 2019). These outcomes demonstrate a potential complementary role of yoga in the management of communicable diseases.

CONCLUSION

There is plenty of evidence that supports that breathing exercises have positive effects on the respiratory system. A simple and cost-effective intervention that could be effortlessly incorporated into a daily routine and has been proven beneficial in subjects across various age groups including the elderly has been Pranayama, a yoga-based respiratory exercise. It has also been documented that yoga training improves the strength of expiratory as well as inspiratory muscles. The beneficial effects of a 6-week course of pranayama on ventilatory lung functions have been reported by Joshi et al. Enhanced ventilatory functions concerning lowered respiratory rate (RR) and increased force vital capacity (FVC), maximum voluntary ventilation (MVV), peak expiratory flow rate (PEFR), and prolongation of breath-holding time have been documented by the authors. To improve the parasympathetic activity in healthy individuals and to strengthen the cardiovascular coupling repeated practice of pranayama is effective. Kapalbhathi, a breathing exercise, is constituted of powerful strokes of exhalation supported with the contraction of diaphragmatic and abdominal muscles followed by passive inhalation. Appropriate training and toning of abdominal and diaphragm muscles is aided by Kapalbhathi. It also assists in cleansing up respiratory passages and the alveoli and also helps in the removal of

secretions from bronchial trees. In competitive swimmers, a collection of yogic breathing techniques enhances pulmonary functions.

The human immune system can play a crucial role in preventing an individual from getting the novel coronavirus infection. Positive effects of yogic practices on the immune system and as well as respiratory system were found in different studies. In the current conditions of COVID-19 public health recommendations promoting physical activity is essential. So, if we successfully incorporate yogic exercises and practices along with social distancing and basic protocols as guided by the World Health Organization like washings hands and staying at home during lockdown then we shall effectively combat this COVID-19 pandemic.

Yoga is not an impossible practice to follow. It offers an abundance of benefits

in turn but asks for very little. A few minutes or hours of yoga in daily life give us a huge amount of physical, mental and spiritual advantages. Even science has ascertained the benefits of yoga now.

In these challenging times the possibilities of falling sick rise sharply. If the immunity levels square down it becomes tough for our body to combat any disease and thus, we must strengthen our defence. Yoga is perhaps the most or at least one of the most beneficial and tried resistant boosters that can help us adapt for a healthier life. It's one of the most ancient arts which not only strengthens the body physically as well as strengthen mental health also by relaxing the mind also. Thus, yoga is a holistic approach for strengthening our physical and mental health. And we must understand that yoga is not an alternative to medicine but a preventive life to stay disease-free.

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